Statistical Inference

Unit 1: Signal, Noise, and What a Test *Really* Tells You

Stat 220 | Statistics for Data Science

What this course is actually about

- Running a statistical test is free. Any library hands you a t-statistic and a p-value in milliseconds, and it will do it just as fast when the answer is garbage.
- The hard part is knowing what the number means *here*, in this messy situation, with this data, for this decision.
- So the question we keep coming back to all semester is not "what is the p-value?" It is: **what would this look like if nothing were going on?**

Let's start by running an experiment on ourselves.

Review: the one-sample t-test, all on one page

You have one sample and a claimed value μ_0. Steps: state the hypotheses, compute t, compare, report.

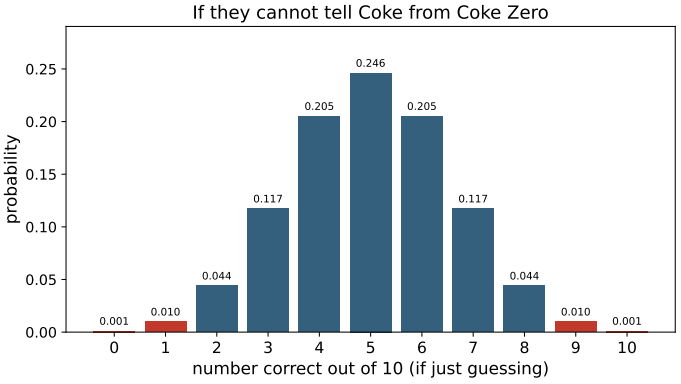
$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \text{df} = n - 1.$$

Alternative	**Use when you care about**	**Reject H_0 when**	**p-value**
$H_1: \mu > \mu_0$ (right-tailed)	only an increase counts	$t > t_{\alpha,\, df}$	$\Pr(T \geq t)$
$H_1: \mu < \mu_0$ (left-tailed)	only a decrease counts	$t < -t_{\alpha,\, df}$	$\Pr(T \leq t)$
$H_1: \mu \neq \mu_0$ (two-tailed)	a change in either direction	$\lvert t \rvert > t_{\alpha/2,\, df}$	$2\Pr(T \geq \lvert t \rvert)$

Trap: choose the side before you see the data

A one-tailed test is easier to pass, so picking the direction after looking at the result doubles your real false-positive rate. If you would have been interested in a move either way, it is two-tailed.

What "just guessing" actually looks like



If they're only guessing, the number they get right follows a simple coin-flip pattern (a Binomial with $n = 10$).

By luck alone:

- ≥ 8 right: **5.5%**
- ≥ 9 right: **1.1%**
- all 10: **0.1%**

A score out in the **red tail** is hard to chalk up to luck.

Where this test came from

- The Coke test has a name. At a 1920s Cambridge tea party, **Muriel Bristol** claimed she could taste whether the milk or the tea was poured into the cup first.
- **Ronald Fisher** did exactly this: assume she's guessing, then ask how surprising her results would be. (He used 8 cups of tea, we used 10 cups of Coke.)
- She got all 8 right, which would happen only about 1.4% of the time by chance.

Principle: the whole idea, in one sentence

A hypothesis test is just the Coke test. Assume "they're only guessing," then measure how surprising the data are. Surprising enough, and you start to doubt the guessing story.

What this unit gives you

Things to know

- sampling variability and the standard error
- the template $(\text{estimate} - \text{null})/\widehat{\text{SE}}$
- the t-statistic and degrees of freedom
- what a p-value does and does not say
- Type I / Type II errors and power
- confidence intervals, and the assumptions underneath it all

Mistakes to stop making

- "p is the probability the null is true"
- confusing statistical with practical significance
- reading "not significant" as "no effect"
- trusting a flashy result from a tiny sample
- ignoring dependence, multiple testing, and peeking

Colors are consistent all semester: **Principle** to keep, **In practice** for a judgment call, **Trap** to avoid.

The one idea behind all of inference

- You have **data**. But you care about **the world**, not the data itself.
- Your data is one random draw from a process that could have come out differently. Run the study again and you'd get different numbers.
- So every statistic you compute (a mean, a difference, a slope) is itself a random quantity with its own distribution.

Principle: estimate vs. parameter

The **parameter** is the true unknown out in the world (μ). The **estimate** ($\bar{x}$) is your data's best guess at it. Inference is the study of how far the estimate might be from the parameter.

A tale of two coffee shops

- Shop A: **4.8** stars from **12** reviews.
- Shop B: **4.5** stars from **2,000** reviews.
- A looks higher, but with only 12 reviews its average is all over the place. A couple of friends of the owner can swing it.
- B's average barely moves no matter who reviews next.

Principle: the moral

A number without its uncertainty is only half a number. "How sure are we?" is the question this whole unit teaches you to answer.

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Now suppose each shop gets one more review, a 0-star:

Shop A: $4.8 \to 4.43$ (a single review drags it down by nearly 0.4). Shop B: $4.5 \to 4.50$ (you can barely see a change).

The point estimate is the easy part

Three ways to report the same A/B test:

"Lift was 2.1%."	A number. Says nothing about whether it is real or how far off it could be.
"Lift 2.1%, $p = 0.03$."	Adds whether it is distinguishable from zero, and nothing else.
"Lift 2.1%, 95% CI 0.2% to 4.0%, on 40k users."	Says how big, how uncertain, and lets the reader decide whether it is worth shipping.

- Any tool will hand you the first one. Producing the third takes knowing what the estimate would have done on a different sample.
- This is the habit that separates people who run analyses from people who are trusted to interpret them, and it costs one extra line of output.

Principle: report the range, not just the middle

A point estimate is a guess. A point estimate with an honest interval is a finding.

Sampling distribution and standard error

- Imagine repeating your study thousands of times. The histogram of all those estimates is the **sampling distribution**.
- Its spread is the **standard error** (SE): the typical distance between your estimate and the truth.

$$\mathrm{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} \qquad \Longrightarrow \qquad \text{quadruple } n \text{ to cut the noise in half.}$$

Principle: SE is the unit of surprise

A difference only means something relative to the SE. "The groups differ by 3" tells you nothing until you know whether 3 is large or small in SE units.

The signal-to-noise template (memorize this)

Almost every test statistic you'll ever meet is the same fraction:

$$\text{test statistic} = \frac{\text{signal}}{\text{noise}} = \frac{\text{estimate} - \text{value under } H_0}{\widehat{\text{SE}}(\text{estimate})}.$$

- **Numerator:** how far your estimate is from what the null predicts (the effect).
- **Denominator:** how much your estimate wobbles from sampling alone.
- A large statistic means the gap is big compared to the wobble.

t-tests, regression slopes, A/B tests: they're all just $(\text{estimate} - \text{null})/\widehat{\text{SE}}$.

Your turn #1

Your turn: which result is stronger evidence?

Two studies each test whether a drug changes blood pressure.

- **Study 1:** estimated drop $= 10$, $\widehat{\mathrm{SE}} = 8$.
- **Study 2:** estimated drop $= 2$, $\widehat{\mathrm{SE}} = 0.5$.

Study 1 has the bigger number. Decide which one is stronger evidence of a real effect, then practice explaining your reasoning to a neighbor for 60 seconds.

Your turn #1: answer

Study 1: $t = \frac{10}{8} = 1.25$ Study 2: $t = \frac{2}{0.5} = 4.0$

- **Study 2** is far stronger evidence, even though its effect estimate is smaller.
- The raw effect (10 vs 2) is a red herring. What matters is the effect compared to its noise.

Principle: the lesson

Never judge an estimate by its size alone. Always ask: compared to what? The answer is its standard error.

What a t-statistic actually is

The one-sample t-statistic is just the template with the mean plugged in:

$$t \;=\; \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \text{df} = n - 1.$$

For comparing two groups (the workhorse of A/B testing):

$$t \;=\; \frac{\bar{x}_A - \bar{x}_B}{\widehat{\mathrm{SE}}(\bar{x}_A - \bar{x}_B)}.$$

- It's a standardized distance: how many standard errors is my estimate from the null value?
- $|t| \approx 2$ is the folklore threshold for "notable," and folklore is where folklore is where the traps live.

Review: the two-sample t-test, all on one page

Two independent groups, and you want to know whether their means differ.

$$t = \frac{\bar{x}_A - \bar{x}_B}{\widehat{\mathrm{SE}}(\bar{x}_A - \bar{x}_B)}, \qquad \widehat{\mathrm{SE}}(\bar{x}_A - \bar{x}_B) = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} \quad \text{(Welch)}.$$

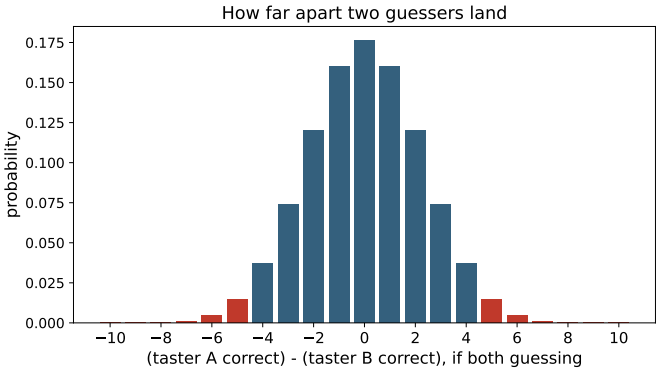
Welch's df comes from a formula nobody memorizes, so let the software report it. Student's pooled version assumes equal variances and is the special case, not the default.

Alternative	**Reject H_0 when**	**p-value**
$H_1:\ \mu_A > \mu_B$	$t > t_{\alpha,\, df}$	$\Pr(T \geq t)$
$H_1:\ \mu_A < \mu_B$	$t < -t_{\alpha,\, df}$	$\Pr(T \leq t)$
$H_1:\ \mu_A \neq \mu_B$	$\lvert t \rvert > t_{\alpha/2,\, df}$	$2\Pr(T \geq \lvert t \rvert)$

Trap: two designs, two different tests

If each subject appears in *both* groups (before and after, left hand and right hand), the observations are not independent. Take the differences per subject and run a **one-sample** test on those. Using the unpaired test there throws away the pairing and most of your power.

How far apart two guessers land



The gap between two scores, if *both* are only guessing:

gap of k or more	chance
≥ 3	0.26
≥ 4	0.12
≥ 5	**0.04**
≥ 6	0.01

A gap of **5 or more** happens only **4%** of the time by luck. "How surprising is the gap?" is exactly the two-sample t question.

The t-test was invented to brew better beer

- **William Gosset** was a chemist at the **Guinness** brewery, working with tiny samples of barley and yeast.
- Normal-based (z) methods were overconfident at small n, so he worked out the heavier-tailed correction we now call the t-distribution.
- Guinness wouldn't let employees publish, so he wrote under the pen name **"Student."** That's why it's *Student's t-test*.

Principle: why it matters

The t-test exists precisely because small samples are dangerous. That caution is built into it, and into half the traps in this unit.

Why "t" and not "z"?

- If we *knew* the true noise σ, the standardized mean would be a standard normal, z.
- We don't. We estimate σ with s from the same small sample, which adds a second source of uncertainty.
- That extra uncertainty gives heavier tails: the t-distribution with $n - 1$ degrees of freedom.

Principle: t turns into z as data pile up

Small n: fat tails, so you need a bigger t to be convinced. Large n: s gets close to σ and t becomes z. Once n is past about 30 to 60, the difference is cosmetic.

Degrees of freedom, intuitively

- Degrees of freedom is how many independent pieces of information are left to estimate variability after you've spent some estimating the mean.
- Spend one on the mean ($\bar{x}$), and $n-1$ are left for s.
- Fewer of them means a noisier estimate of the noise, which means fatter tails and more caution.

Trap: "df is just $n-1$, who cares"

With small samples, ignoring df makes you overconfident: you treat a wobbly noise estimate as if it were exact. The whole point of the t-distribution is to stop you.

The idea so far

- Your data is one draw from a process that could have come out differently, so every number you compute has a distribution.
- The width of that distribution is the **standard error**.
- Every test statistic is $\dfrac{\text{estimate} - \text{what the null predicts}}{\text{standard error}}$, which asks: is this gap big *compared to how much it naturally wobbles*?

Principle: the question now

Now we turn that ratio into a probability, and then spend most of this unit on what that probability does not mean.

The p-value, stated carefully

Principle: say it this way, every time

The p-value is the probability of a test statistic **at least as extreme** as the one you observed, **assuming the null hypothesis is true**.

$$p \;=\; \Pr\big(\,|T| \geq |t_{\text{obs}}| \;\big|\; H_0 \text{ true}\,\big).$$

- **You already computed one.** In the Coke test, "$\Pr(\geq 8 \text{ right} \mid \text{guessing}) = 5.5\%$" is a p-value.
- A small p says "data like mine would be surprising if the null were true." That's all it says.

What a p-value is *not*

Trap: the five classic misreadings

1. It is not the probability the null is true. ($p \neq \Pr(H_0 \mid \text{data})$.)
2. It is not the probability your result happened "by chance."
3. $1 - p$ is not the probability the alternative is true.
4. A smaller p does not mean a bigger or more important effect.
5. $p > 0.05$ does not mean "no effect." It means "not enough evidence here."

What p-values do when nothing is going on

Run a study where the two groups are genuinely identical, then do it 5,000 more times. Plot the p-values you get.

- The histogram comes out **flat**. Under the null the p-value is **uniform** on $[0, 1]$: every value is equally likely, and a p of 0.03 is exactly as likely as a p of 0.83.
- So $p < 0.05$ happens in **5% of studies where nothing is happening**, by construction. That is not a surprising event, it is the definition.

Principle: the 5% is a budget you chose, not evidence you found

α is the false-alarm rate you agreed to tolerate before collecting data. A single significant result is one draw from that budget, which is why twenty tests on nothing produce about one "discovery."

You will build this histogram yourself in the code companion. It is the fastest way to stop over-reading a small p.

What actually drives a p-value

Rewrite the two-group t and you can see what's inside it:

$$t \approx \underbrace{\frac{\text{effect size}}{\sigma}}_{\text{standardized signal}} \times \underbrace{\sqrt{\frac{n}{2}}}_{\text{sample size}} .$$

- A p-value tangles three things together: the true effect, the noise, and the sample size.
- Hold the effect fixed and grow n: $|t|$ grows like $\sqrt{n}$ and p heads toward 0.

Principle: the headline consequence

With enough data, almost any effect, however trivial, turns "statistically significant."

Scenario 1: a tiny p-value with a trivial effect

Setup. $n = 5{,}000{,}000$ users. Mean session time differs between two designs by **0.4 seconds**, $p < 10^{-8}$.

- The p-value is tiny because $|t|$ grows with $\sqrt{n}$, not because the effect is large.
- The effect is 0.4s (trivial?), the CI is maybe $[0.36, 0.44]$s (precise and small), and the decision really comes down to business value.

In practice: how to answer this

"It's real, but probably irrelevant. At 5M users we can resolve effects far smaller than anything worth shipping. Show me the effect size and the CI. Significance alone tells us little here."

Scenario 2: twenty tests, one "winner"

Setup. A team tests 20 jelly-bean colors against acne. Nineteen show nothing, then *green* hits $p = 0.04$, and the headline writes itself.

- Under a true null, each test fires falsely 5% of the time. Across 20 tests, the chance of at least one false positive is $1 - 0.95^{20} \approx 64\%$.
- So a lone $p = 0.04$ out of 20 is about what pure noise produces.

Trap: multiplicity and p-hacking

Every extra test, segment, or peek inflates the false-positive rate. Correct for it (Bonferroni, FDR) or decide on the one test that matters ahead of time.

Your turn #2

Your turn: practice explaining it

A clinical trial reports $p = 0.04$. A news article writes:

"There is only a 4% chance the drug doesn't work."

True or false? Then turn to a neighbor and practice explaining, in your own words, what $p = 0.04$ really says.

Your turn #2: answer

False.

- $p = 0.04$ is computed *assuming the drug does nothing*. It can't also be the probability that the drug does nothing. That would be circular.
- The right reading: if the drug truly did nothing, we'd see data this striking only about 4% of the time.
- The probability the drug works *given* the data ($\Pr(H_1 \mid \text{data})$) is a different quantity, and it needs a prior. That is Bayes, and Unit 11.

The hypothesis-testing setup

- **Null H_0:** the boring default, like "no difference" or "slope is 0."
- **Alternative H_1:** what you'd act on, one-sided or two-sided.
- **Level α:** the false-alarm risk you accept before seeing data, often 0.05, which is a convention and not a law.
- **Decision:** reject H_0 if $p < \alpha$.

	H_0 **true**	H_1 **true**
Reject H_0	Type I error (α)	Correct (power $= 1 - \beta$)
Fail to reject	Correct	Type II error (β)

Principle: what each error costs

Type I: you ship a change that does nothing. Type II: you kill a good idea. You choose α outright. Type II is governed by power, which you do not choose directly.

Power

Principle: definition

Power $= 1 - \beta = \Pr(\text{reject } H_0 \mid H_1 \text{ true})$: the chance your test *catches* an effect that is really there.

Power goes up with:

- a larger true effect
- a larger sample size n
- smaller noise σ.

Trap: low power distorts, it doesn't just miss (the "winner's curse")

When power is low, the only estimates that clear significance are the ones that got lucky and large, so "significant" effects come out systematically overstated. That's why splashy small-sample findings so often fail to replicate.

Your turn #3

Your turn: is the colleague right?

A new onboarding flow is tested on $n = 10$ users. Result: $p = 0.40$. A colleague concludes: *"Proven. The new flow makes no difference."* Do you agree? What's the first question you ask?

Your turn #3: answer

Answer to Your turn #3: no. "Not detected" is not the same as "not there."

- With $n = 10$ the study is badly underpowered: it probably couldn't detect even a large, real effect, so $p = 0.40$ tells you very little.
- First question: what effect *could* this study have detected? Then read the CI. Something like $[-3\%, +12\%]$ has not ruled out a big positive effect.

Principle: absence of evidence is not evidence of absence

A non-significant result with a wide CI is inconclusive, not negative. The honest answer is "we need more data," not "no effect."

Confidence intervals: usually the better answer

- A 95% CI is the set of null values your data would not reject: estimate $\pm$ (critical t) $\times$ $\widehat{\mathrm{SE}}$.
- The connection to tests: a two-sided test rejects μ_0 at level α exactly when the $(1-\alpha)$ CI leaves μ_0 out.

Principle: why a CI usually beats a p-value

The CI shows the effect's size and its uncertainty in a single object. "Lift $+2\%$, 95% CI $[-1\%, +5\%]$" says far more than "$p = 0.21$."

Statistical vs. practical significance

Statistical significance

- "Distinguishable from the null."
- Driven by effect *and* n.
- A p-value question.

Practical significance

- "Big enough to act on."
- Driven by effect size and cost.
- A *business* question.

In practice: the question behind half of all A/B tests

"2M users, the new button raised clicks by 0.03%, $p < 10^{-6}$. Ship it?" The statistics are settled. The real decision is whether 0.03% is worth the engineering and the risk.

Scenario 4: the A/B test that was "peeked"

Setup. An experiment runs on a live dashboard, and the team ships the moment it crosses $p < 0.05$.

- Testing over and over as data arrive (optional stopping) is multiplicity in disguise, and the true false-positive rate climbs well above 5%.
- A pure-noise experiment, watched long enough, will eventually cross 0.05 on its own.

You may be asked: a standard screening question

"Peeking breaks the nominal α. Either fix the sample size in advance, or use methods built for monitoring, like sequential or always-valid tests. And separately: is the lift worth shipping?"

What a t-test quietly assumes

Each assumption is easiest to remember as the thing that breaks it.

1. **Independence.** Every row is its own piece of information. *Broken by:* 300 students who sat in 10 study groups and compared answers. That is closer to 10 opinions than 300.
2. **The estimate is roughly normal**, which is not the data being normal. *Broken by:* 12 customers with wildly skewed spending. At 500 the average is fine even though no individual is.
3. **The groups share a spread**, or you use Welch, which does not need it. *Broken by:* 2,000 long-time users who behave alike against 80 new ones who do not.
4. **No single point runs the show.** *Broken by:* 18 purchases, one of them a $45,000 tractor.

Principle: which one to worry about first

Independence by a wide margin, then outliers and skew at small n, then unequal variance, and last the shape of the raw data.

What each failure actually does to you

What went wrong	Why it hurts	What you see
Rows are related to each other	Asking someone and then their roommate is not two opinions.	The SE divides by a $\sqrt{n}$ that is too big, so everything looks more certain than it is.
One or two extreme values	That point moves the mean *and* inflates s.	A result that flips when you delete one row.
Heavy skew, small n	The average of a few skewed numbers is still skewed.	A p-value that is simply wrong, either way.
Different spreads, different group sizes	Pooling forces one spread on two groups that do not share one.	Student's t claims more certainty than it has.

Trap: the one that never announces itself

The other three leave fingerprints in a plot. Dependence does not. Only knowing how the data was collected tells you.

Skewed data: three honest options, three different answers

Rent, income, and revenue are all right-skewed. Three ways to run the comparison, and they are not three attempts at the same number.

What you do	What you are estimating	What it costs
t-test on the raw values	The difference in **mean** rent, in dollars.	The mean is dragged by a few extreme listings, and at small n the p-value is not trustworthy.
t-test on $\log(\text{rent})$	The difference in mean *log* rent, which back-transforms to a **ratio**, not dollars.	If someone asked "how many more dollars," you did not answer it.
t-test dropping the extreme 1%	The difference in mean rent **among typical listings**.	You changed the population, and must say so.

Principle: pick the target before you pick the fix

All three are defensible and answer different questions. The mistake is running all three and reporting whichever gave the smallest p-value.

A checklist for any result you're handed

Run this on every test statistic someone shows you:

1. **Effect size:** how big is it, in units anyone cares about?
2. **Uncertainty:** what's the SE or confidence interval?
3. n **and power:** could this n detect a meaningful effect? a meaningless one?
4. **Assumptions:** independence? outliers? variance? design?
5. **Multiplicity:** how many tests or looks produced this number?
6. **Decision:** given the costs, what would you actually do?

Your turn #4

Your turn: same n, same formula, same answer?

Two researchers each want the average resting heart rate of adults, and each collects **1,000 readings**.

- **A** measures 1,000 different people, once each.
- **B** measures **5** people, 200 times each.

Both report a mean of 68 bpm with $\widehat{\text{SE}} = s/\sqrt{1000}$, and both get a very tight interval.

With a neighbor: are both of those standard errors believable? If not, whose, and what is that researcher's real sample size?

Your turn #4: answer

A is fine. B is not, and B's real sample size is much closer to 5 than to 1,000.

- B's 200th reading from person 3 says almost nothing new *about adults*. It says something about person 3, who is already in the data.
- But the formula does not know that. Every extra row divides $\widehat{\mathrm{SE}} = s/\sqrt{n}$ by a little more, **as if new information had arrived**. Almost none did.
- Take the extreme case where a person's repeats are identical. Then B really has 5 numbers, and the honest denominator is $\sqrt{5}$, not $\sqrt{1000}$. The reported SE is too small by a factor of $\sqrt{1000/5} = \sqrt{200} \approx \mathbf{14}$, so t is about **14 times too big**.
- Real repeats are not identical, so the truth sits between the two. It sits much nearer the bad end.

Principle: count independent observations, not rows

More rows only buy precision when each one is a fresh draw from the population you care about.

What to do about it, with what you already have

- **Ask the question first:** what is *one* independent observation here? A person, a store, a week, a classroom? That answer, not the row count, is your n.
- **Then collapse to it.** Average each person's 200 readings into one number, and run the analysis on those 5 values. You throw away 995 rows and gain an honest answer.
- The same move handles the business version: 400,000 sessions from 12,000 users is far closer to 12,000 than to 400,000. Average to the user, then compare.

Trap: a giant t is not reassurance

When the unit is wrong, a huge test statistic is the most dangerous output you can get. A t of 31 is then telling you about your denominator, not about the world.

Scenario 5: significant, but small and skewed

Setup. $n = 18$ per group, revenue data (heavy right skew, one "whale"), $p = 0.02$.

- Two red flags collide: a small n (the CLT hasn't rescued normality) and a single outlier that can manufacture, or hide, significance.
- **Quick diagnostic:** compare the mean to the median. On symmetric data they nearly agree. When the mean sits far above the median, a tail is pulling it, and the mean is no longer describing a typical case.
- Pressure-test it: does the conclusion survive if you drop that one observation? What if you compare medians instead of means, or analyze on a log scale?

In practice: how to answer this

"I don't trust this p yet. With $n = 18$ and a heavy tail, one observation may be doing all the work. I'd show it three ways: raw, with that observation dropped, and on the log scale."

Scenario 6: unequal variances or paired data

Setup. Group A ($n = 2000$, low variance) vs. Group B ($n = 80$, high variance).

- Unequal variances together with unequal n is exactly where Student's t misfires. Use **Welch's** t. It is free and it fixes the problem.
- Different design: was each user measured before and after? Those two numbers aren't independent, so use a **paired** test on the differences, or you throw away power.

In practice: how to answer this

"First I'd ask about the design: independent or paired, and are the variances comparable? That choice, Welch vs. Student and paired vs. unpaired, changes the answer more than the data do."

Running a two-group comparison, start to finish

1. **Write down H_0, H_1, and α before you look.** One-sided or two-sided is decided here, not later.
2. **Check the design.** Independent groups, or the same subjects measured twice? Paired data goes to a one-sample test on the differences.
3. **Plot both groups** before any test. Look for skew, outliers, and whether the groups are even comparable.
4. **Report n, mean, and SD** for each group.
5. **Run Welch's test**, which does not assume equal variances: `stats.ttest_ind(a, b, equal_var=False)`.
6. **Compute the CI for the difference**, $(\bar{x}_A - \bar{x}_B) \pm t^* \widehat{\mathrm{SE}}$, because that is the number you will actually report.
7. **Report all four:** the effect size in real units, the interval, the p-value, and n.

The traps

Trap: the ones that bite most often

- "p is the probability the null is true" (it isn't).
- "Smaller p means a bigger effect" (no, p confounds effect, noise, and n).
- Treating statistical significance as practical importance (different questions).
- Reading non-significant as no effect (could be low power, so check the CI).
- Trusting a significant estimate from a tiny sample (winner's curse).
- Ignoring multiplicity, peeking, and p-hacking.
- Treating dependent data as independent (silent, deadly, inflates t).
- Worshiping 0.05 as a cliff instead of reading p as continuous evidence.

Sixteen practice questions on exactly these, with worked answers, are in `Practice_Inference.pdf`.

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
One-sample t-test against a claimed value	`stats.ttest_1samp`
Two independent groups (Welch, the default choice)	`stats.ttest_ind(..., equal_var=False)`
Before-and-after on the same subjects	`stats.ttest_rel`
Confidence interval for a mean or a difference	`stats.t.ppf` with $s/\sqrt{n}$
Critical value or a p-value from t	`stats.t.ppf`, `stats.t.sf`
Power, and the n you would need	`statsmodels.stats.power.TTestIndPower`
Simulate the null to check any of the above	`rng.normal`, then loop

If you cannot say what the output means, the fact that you produced it is worth nothing.

Where we go next

- You now have the template: an estimate, its standard error, and the discipline to ask what chance alone would have produced.
- **Unit 2** lays out what models are available. **Unit 3** then points this template at a relationship instead of a difference: the estimate becomes a *slope*, and almost everything from this unit transfers unchanged.
- The rest of Part I builds and stress-tests models. Part II, starting at Unit 6, goes back and earns the machinery we are about to use on credit.

Principle: the habit to keep

Before you explain a result, work out what pure chance would have produced. Most of this course is that one habit, applied to harder situations.